

Stable and Reversible Triphenylphosphine-Based n-Type Doping Technique for Molybdenum Disulfide (MoS₂)

Keun Heo,^{†,§} Seo-Hyeon Jo,^{†,§,||} Jaewoo Shim,[†] Dong-Ho Kang,[†] Jeong-Hoon Kim,[†] and Jin-Hong Park^{*,†,‡,||}

[†]Department of Electrical and Computer Engineering and [‡]SKKU Advanced Institute of Nano Technology (SAINT), Sungkyunkwan University, Suwon 16419, Korea

S Supporting Information

ABSTRACT: A highly stable and reversible n-type doping technique for molybdenum disulfide (MoS₂) transistors and photodetectors is developed in this study. This doping technique is based on triphenylphosphine (PPh₃) and significantly improves the performance of MoS₂ transistor and photodetector devices in terms of the on/off-current ratio ($8.72 \times 10^4 \rightarrow 8.70 \times 10^5$), mobility ($12.1 \rightarrow 241 \text{ cm}^2/\text{V}\cdot\text{s}$), and photoresponsivity (R) ($2.77 \times 10^3 \rightarrow 3.92 \times 10^5 \text{ A/W}$). The range of doping concentrations is broadly distributed between 1.56×10^{11} and $9.75 \times 10^{12} \text{ cm}^{-2}$ and is easily controlled by adjusting the temperature at which the PPh₃ layer is formed. In addition, this doping technique provides two interesting properties that have not been reported for previous molecular doping techniques: (i) high stability leading to small variations in device performance after exposure to air for 14 days (on-current: 1.34% and photoresponsivity: 1.58%) and (ii) reversibility enabling the repetitive formation and removal of PPh₃ molecules (doping and dedoping).

KEYWORDS: MoS₂, transistor, photodetector, PPh₃, air-stable, reversible doping

INTRODUCTION

Currently, various electronic and optoelectronic devices based on transition-metal dichalcogenides (TMDs) with a two-dimensional layered structure are being actively researched.^{1–7} In this circumstance, there is a continuous demand for the development of doping techniques that can improve device performance by precisely controlling the carrier concentration.^{8–12} In electronic devices, the doping technique can minimize inefficient voltage loss and subsequently improve the on-state current by reducing the effective barrier and contact resistance height at the metal/TMD junction.^{8,10} Additionally, the doping enables control of the recombination rate of photoexcited carriers, and this enhances the performance of optoelectronic devices.^{10,12} The carrier recombination rate is largely influenced by the number of trions formed from excitons in accordance with doping concentration.^{10,13} Current TMD doping techniques are categorized into the following two methods: (i) atom-substitutional doping^{8,9,14–17} and (ii) molecular doping.^{10,11,18,19} Substitutional doping replaces transition-metal (or chalcogen) atoms with dopants in the TMD growth step and exhibits the advantage of not harming the crystallinity of TMDs.^{8,9,14–17} However, it is complex and difficult to accurately control the relative amounts of three different atoms (dopant, transition metal, and chalcogen atoms), and hence, precise doping control has not yet been reported for TMD materials doped by substitutional doping techniques.^{8,9,14–17} In molecular doping, TMD materials are

doped by the direct transfer of carriers from dopant molecules to TMDs as determined by the difference between the reduction potential of the dopant molecules and the chemical potential of the TMDs. Because the surface molecular layers exhibit various dipole moments and the amount of molecules can be adjusted, it is possible to precisely control the carrier concentration of TMDs.^{10,11,18,19} However, most molecular doping techniques suffer from relatively poor stability^{10,19} compared to substitutional doping.^{9,14}

This study establishes a highly stable and reversible n-doping technique for molybdenum disulfide (MoS₂) that is based on charge transfer from triphenylphosphine (PPh₃). This n-doping method enables a high on/off-current ratio (exceeding 10^5), high mobility (approximately $241 \text{ cm}^2/\text{V}\cdot\text{s}$), and high R (exceeding 10^5 A/W) in the MoS₂ transistor and photodetector devices. First, n-doping by a PPh₃ layer on MoS₂ is analyzed with Raman spectroscopy and X-ray photoelectron spectroscopy (XPS), and the doping concentration is adjusted by controlling the annealing condition. The effects of PPh₃ doping on MoS₂-based devices are also investigated in terms of the resulting electronic (threshold voltage, drain current level, field-effect mobility, and carrier concentration) and optoelectronic (photoresponsivity and photoresponse time) device

Received: April 25, 2018

Accepted: September 3, 2018

Published: September 17, 2018

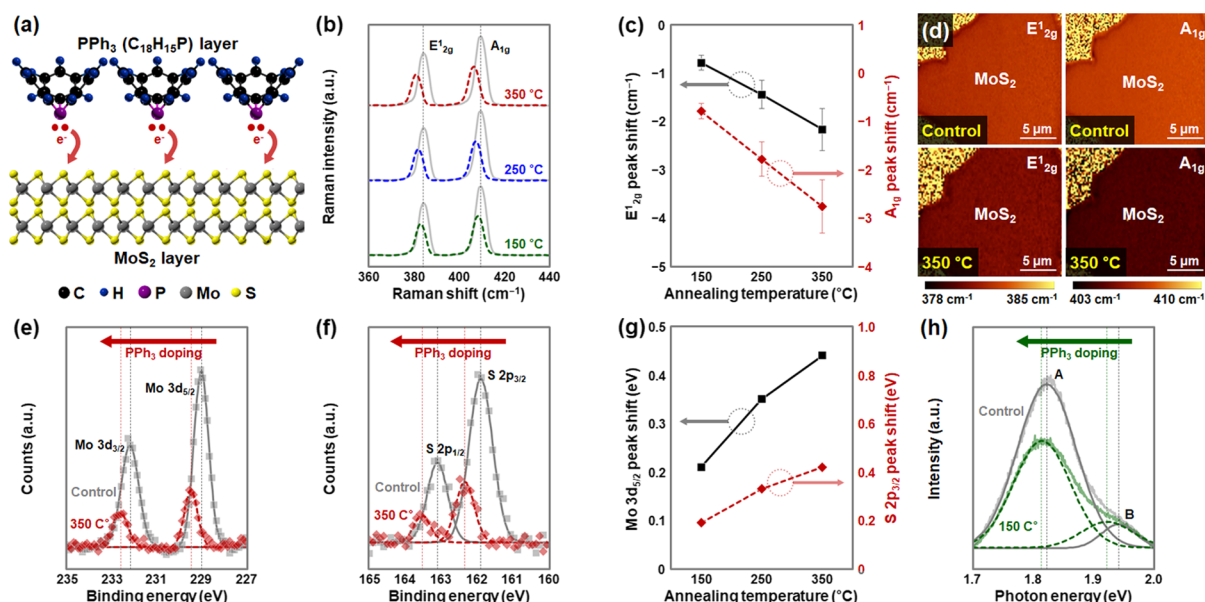


Figure 1. Graphical description and spectroscopic analysis of PPh₃-doped MoS₂ at varying annealing temperatures: (a) schematic diagram of the PPh₃ doping mechanism at the PPh₃/MoS₂ interface; (b) Raman spectra of pristine (solid gray line) and PPh₃-doped (dotted color lines) MoS₂ at different annealing temperatures (150, 250, and 350 °C); (c) extracted Raman peak shift values as a function of the PPh₃ formation temperatures; (d) mapping images corresponding to E_{2g}¹ and A_{1g} peaks before and after PPh₃ doping at 350 °C; (e) XPS spectra of Mo 3d peaks; (f) XPS spectra of S 2p peaks; (g) extracted peak shifts for Mo 3d and S 2p at varying annealing temperatures; (h) PL analysis of the pristine (solid gray line) and PPh₃-doped (dotted green line) MoS₂.

performance. Subsequently, the stability of the PPh₃ doping effect is evaluated by monitoring the device parameters in air for 336 h. The reversibility of this PPh₃ doping technique is finally demonstrated by repeatedly forming and removing the PPh₃ molecular layer.

RESULTS AND DISCUSSION

To investigate the effects of PPh₃ doping on MoS₂ under varying annealing conditions, Raman spectroscopy and XPS analyses were performed before and after applying the PPh₃ layer, where the PPh₃-coated MoS₂ samples were annealed at three different temperatures (150, 250, and 350 °C). Figure 1a shows an illustration of the n-doping mechanism that occurs at the PPh₃/MoS₂ interface. After the surface of MoS₂ is coated with PPh₃, the electron lone pairs of the phosphorus atoms in the PPh₃ molecules transfer negative charges (electrons) to MoS₂, thereby causing the n-doping phenomenon. As expected, the two Raman peaks observed at 384 and 409 cm⁻¹ in the spectra shown in Figure 1b, which correspond to the E_{2g}¹ and A_{1g} phonon modes of MoS₂, were red shifted (dotted color lines).²⁰ This result is consistent with previous Raman studies, in which n-doping of MoS₂ with APTES,¹⁰ BV,¹⁴ ATO,²¹ DNA,²² and PSG²³ also resulted in a red shift in the MoS₂ Raman peaks. To analyze the peak shift tendency at varying annealing temperatures, we extracted the peak shift value from each Raman peak (E_{2g}¹ and A_{1g}) and plotted it as a function of the annealing temperature (Figure 1c). We prepared five samples and randomly selected three points on each sample for Raman measurement so that the data points and error bars in Figure 1c were obtained by considering 15 different Raman spectra. At each annealing condition, 10 different samples were prepared and the Raman measurement was performed at five different points on each sample. With increases in the annealing temperature during the PPh₃ formation, the shift of the Raman peaks increased from

−0.78 to −2.16 cm⁻¹ for the E_{2g}¹ peak and from −0.78 to −2.75 cm⁻¹ for the A_{1g} peak. These phenomena indicate that PPh₃ causes the softening of MoS₂ vibrational modes and consequently causes stronger n-doping effects as the PPh₃ annealing temperature increases.²⁴ Spectral mapping of the Raman peak shift on the surface of MoS₂ was performed to confirm the uniformity of the PPh₃ doping. Figure 1d shows the mapping images corresponding to the E_{2g}¹ and A_{1g} peaks before and after PPh₃ doping at 350 °C. Compared to the images obtained before doping, those obtained after the PPh₃ doping appeared uniformly darkened over the entire area, thereby indicating that the MoS₂ layer was uniformly n-doped by PPh₃. This uniform n-doping was quantitatively confirmed by the standard deviation values extracted from the histograms of the mapping images that are explained in Supporting Information Figure S1.

Subsequently, XPS analyses of MoS₂ before and after PPh₃ doping were conducted to support the previous Raman analysis of the PPh₃ effects. As shown in Figure 1e,f, the Mo 3d and S 2p peaks, which are associated with the bonding configurations of the Mo and S atoms in MoS₂, were positively shifted by 0.44 and 0.42 eV, respectively, after PPh₃ doping at 350 °C. This result indicated that the Fermi level of MoS₂ moved up toward the edge of the conduction band, and this confirmed the n-doping of MoS₂.^{25,26} Previous XPS studies on the n-doping of MoS₂ using dopant molecules, such as Cs₂CO₃,¹² K,²⁶ (2-Fc-DMBI)₂,²⁷ and 2-Fc-DMBI-H,²⁷ also reported a similar up-shifting phenomenon in the binding energy of peaks related to MoS₂. To investigate the dependence of the peak shifts on the annealing temperature, peak shift values were extracted from the XPS peaks (Mo 3d_{5/2} and S 2p_{3/2}) obtained for samples annealed at different temperatures. As shown in Figure 1g, as the annealing temperature increases, the Mo 3d_{5/2} peak is shifted from 0.21 to 0.44 eV and the S 2p_{3/2} peak is shifted from 0.19 to

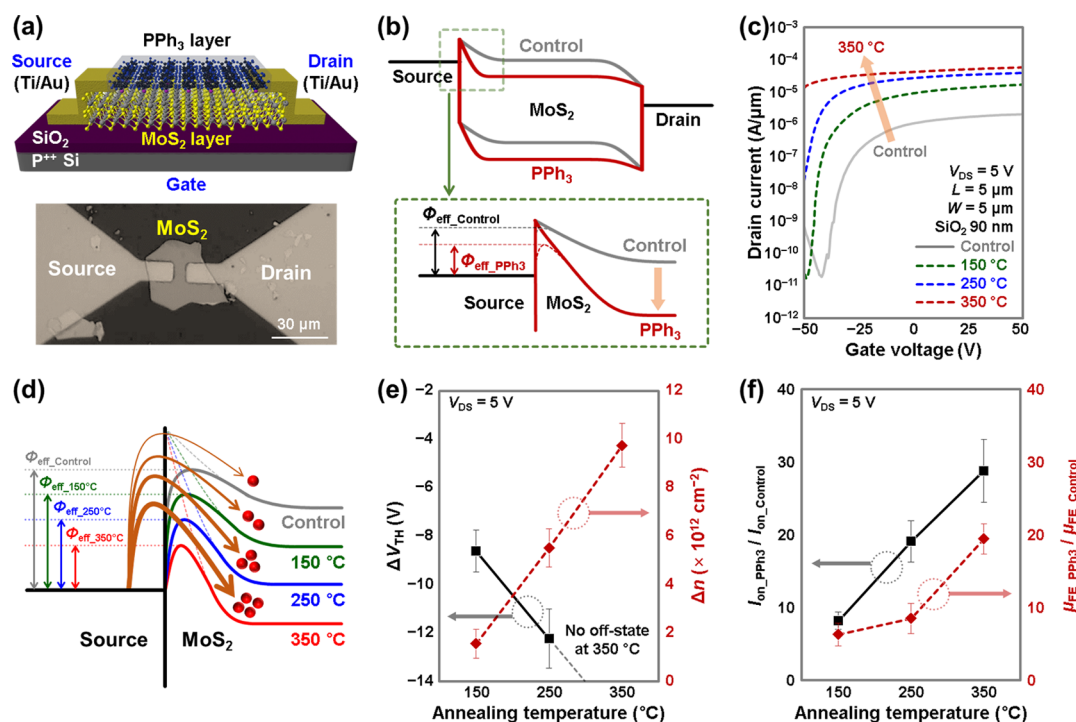


Figure 2. Electrical characterization of PPh₃-doped MoS₂ photodetectors: (a) schematic diagram and optical image of the PPh₃-doped MoS₂ transistor; (b) energy band diagram of the source–TMD–drain junction before and after PPh₃ doping; (c) I_D – V_G characteristics of the control (solid gray line) and PPh₃-doped (dotted color lines) MoS₂ devices activated at three different temperatures (150, 250, and 350 °C); (d) lowering of the effective electron barrier height at the source-TMD junction as a function of annealing temperature; (e) threshold voltage shift ($\Delta V_{TH} = V_{TH_PPh_3} - V_{TH_Control}$) and carrier concentration difference ($\Delta n = n_{PPh_3} - n_{Control}$); (f) extracted on-current ratio ($I_{on\ ratio} = I_{on_PPh_3} / I_{on_Control}$) and field-effect mobility ratio ($\mu_{FE\ ratio} = \mu_{FE_PPh_3} / \mu_{FE_Control}$) extracted at $V_{GS} = -50$ V.

0.42 eV. This trend is in agreement with the Raman results shown in Figure 1b,c, in which the peak shift also becomes more evident as the annealing temperature increases. The PPh₃ layers formed at different temperatures were quantitatively analyzed with the XPS spectra for the P 2p and C 1s peaks that are related to the bonding configurations of constituent atoms inside the PPh₃ molecules. Details of the XPS spectra are provided in Supporting Information Figure S2. An increase in the phosphorus and carbon density of MoS₂ was observed with increases in the annealing temperature. This implies that the PPh₃ layer became denser with increases in the annealing temperature and subsequently strengthened the electron charge-transfer phenomenon. It is because (i) a PPh₃ molecule (see below) in PPh₃ has one pair of lone electron being transferred to the TMD layer and (ii) more PPh₃ molecules are expected to exist in the denser PPh₃ layer. Additionally, photoluminescence (PL) analysis was performed to investigate the change in the optical characteristics of MoS₂ after PPh₃ doping. The PL peaks observed at 1.82 and 1.94 eV (Figure 1h) are related to the A and B characteristic peaks of MoS₂, respectively, and exhibited a red shift.^{10,12} This is potentially due to the formation of negatively charged trions from excitons as a result of the increase in the electron concentration in MoS₂ after PPh₃ doping.^{10,12} Because the relative contribution of the trion peak increased when compared to the exciton peak, red shifting typically occurred in the PL peaks. Details of the PL analysis are specifically described in Supporting Information Figure S3. This is also consistent with the results of previous studies on the tunable PL characteristics of MoS₂ obtained with electrostatic and chemical doping in which n-doped MoS₂ caused a red shift in the peak positions.^{10,12}

To investigate the effects of PPh₃ doping on the performance of MoS₂ electronic devices, back-gated field-effect transistors (FETs) were fabricated and electrical measurements were obtained before and after PPh₃ doping. Five different device samples were prepared and measured at each annealing condition. Figure 2a shows schematic and optical images of the MoS₂ FET device (the channel length and width are both 5 μ m). After applying PPh₃ doping to the MoS₂ device, the energy band of the MoS₂ shifts downward as shown in Figure 2b because the n-type doping transfers electrons to MoS₂. Consequently, the effective barrier height (Φ_{eff}) and the tunneling barrier width for the electrons decrease at the source/MoS₂ junction, thereby increasing the injection probability for electron carriers from the source to MoS₂. The I_D – V_G curves (at $V_{DS} = 5$ V) shown in Figure 2c indicate that the on-current (I_{on}) increased and the threshold voltage (V_{TH}) decreased after PPh₃ doping. The aforementioned effects of n-doping on the MoS₂ devices became stronger with increases in the annealing temperature for the formation of PPh₃ on MoS₂. This was because there was an increase in the number of electrons transferred to MoS₂ from the relatively dense PPh₃ layer formed at higher temperatures. The lower electron effective barrier height and the thinner tunneling barrier width are predicted at the source/MoS₂ junction as the annealing temperature increases, resulting in a higher injection probability for electrons to MoS₂ (Figure 2d). Therefore, the extracted ΔV_{TH} decreased from –8.62 V to less than –12.2 V when the doping concentration ($\Delta n = n_{PPh_3} - n_{Control}$) increased from 1.56×10^{12} to 9.75×10^{12} cm^{–2}, as shown in Figure 2e. It should be noted that it was difficult to extract the V_{TH} value for the samples annealed at 350 °C because the

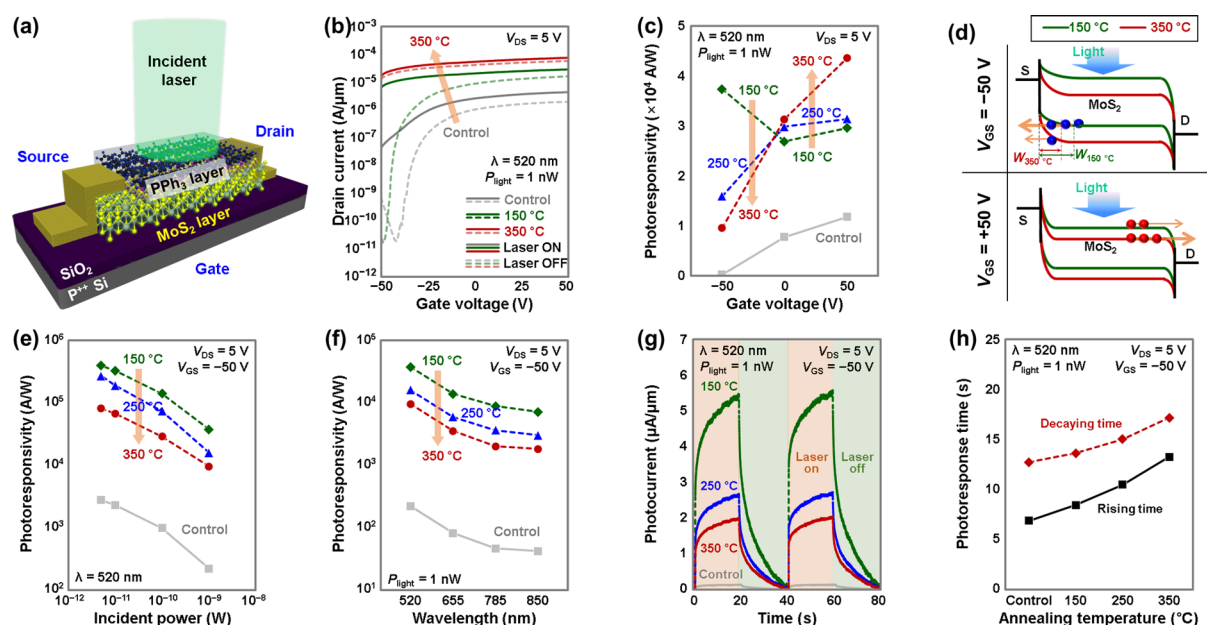


Figure 3. Photoresponse characteristics of PPh₃-doped MoS₂ photodetectors: (a) schematic diagram of the PPh₃-doped MoS₂ photodetector and (b) photocurrent response. Solid and dashed lines represent I_D – V_G curves (at $V_{DS} = 5$ V) measured under laser on and off conditions, respectively; (c) photoresponsivity as a function of gate bias, which were obtained in the MoS₂ photodetectors activated at three different temperatures (150, 250, and 350 °C); (d) energy band diagrams of source–MoS₂–drain junctions under gate biases of (top) +50 V and (bottom) –50 V in which the MoS₂ is doped by PPh₃ at 150 and 350 °C; (e) photoresponsivity as a function of incident laser power and (f) wavelength, which were measured in the control (solid gray line) and PPh₃-doped (dotted color lines) MoS₂ photodetectors activated at three temperatures; (g) temporal photoresponse characteristics and (h) extracted rising and decaying times for MoS₂ photodetectors formed at the three activation temperatures.

MoS₂ device was degenerately doped (refer to the I_D – V_G curve in Figure 2c). We also note that 10 different I_D – V_G curves were measured on different devices to obtain the sufficient data for the ΔV_{TH} , Δn , I_{on} ratio, and μ_{FE} ratio. The Δn values in Figure 2e were obtained over a wide range from nondegenerate to degenerate doping levels, and this covers the doping concentrations previously reported for CH₄-SAM nondegenerate n-doping (8.0×10^{11} cm^{–2}),²⁸ SiN_x nondegenerate n-doping (2.0×10^{12} cm^{–2}),²⁹ APTMS nondegenerate n-doping (7.2×10^{12} cm^{–2}),³⁰ K degenerate n-doping (1.0×10^{13} cm^{–2}),²⁶ and BV degenerate n-doping (1.2×10^{13} cm^{–2}).¹⁴ Additionally, the I_{on} and field-effect mobility (μ_{FE}) increased from 8.17 to 28.8 (I_{on} ratio) and from 7.11 to 19.9 (μ_{FE} ratio) with increases in the annealing temperature, as shown in Figure 2f. The μ_{FE} extracted at $V_{DS} = 0.1$ V also improved from 4.67 to 70.82 cm²/V·s as the doping temperature increased, but the values were slightly lower than those obtained at $V_{DS} = 5$ V (Supporting Information Figure S4 and Table S1). We also investigated the contact resistance (R_c) at various doping conditions. The R_c values decreased from 2.82 to 0.24 k Ω with increasing doping temperature (Supporting Information Figure S5). This inverse relationship is attributed to the reduction of the effective barrier height with increase of PPh₃ forming temperature.

The effects of PPh₃ doping on the MoS₂ photodetector performance were also investigated. As shown in Figure 3a, a light was exposed to the channel region between the source and drain electrodes before and after PPh₃ doping of the MoS₂ photodetector. After the PPh₃ doping, a higher electric field formed in the channel region is predicted to improve the collection of photocarriers because the contact resistance between Ti and MoS₂ decreases owing to reduction in the effective electron barrier height (Figure 3b). In particular, under different gate bias conditions, the photoresponsivity

varied in opposite directions with the formation temperature of PPh₃ (Figure 3c). The upper diagram in Figure 3d shows the energy band structures of Ti–MoS₂–Ti junctions doped with PPh₃ layers formed at 150 or 350 °C and illuminated under a gate bias of –50 V. As shown in the figure, photogenerated hole carriers primarily contribute to the photoresponsivity. Specifically, the device with PPh₃ formed at 350 °C exhibited a worse photocarrier collection than the device doped at 150 °C because the depletion width became narrower at the source–channel junction and resulted in a lower photoresponsivity. At a positive gate bias of +50 V, electron carriers are accumulated in the conduction band states. Thus, the depletion widths at Ti–MoS₂ junctions doped at 150 and 350 °C are expected to be extremely short (few nanometers) and not much different. Photoresponsivity is also influenced by carrier lifetime related linearly to tightly bounded trions; therefore, in this case, tightly bounded trions potentially play a major role in photoresponsivity.^{10,12} When the electron concentration increases due to the PPh₃ n-doping formed at higher annealing temperature, the number of tightly bound trions increases and the recombination rate of excitons decreases. The reduced PL peak intensities of the PPh₃-doped MoS₂ in Figure 1h also indicate a decrease in the recombination probability of carriers. This reduction in the recombination rate increased the photocarrier lifetime of the MoS₂ photodetectors, resulting in higher photoresponsivity.

In addition, temperature-dependent photoresponse characteristics were investigated by varying incident optical powers and wavelengths (Figure 3e,f). With respect to the gate bias condition of –50 V in which a sufficiently low dark current was achieved, a maximum photoresponsivity of 3.92×10^5 A/W was obtained for the device doped at 150 °C and exposed under a 5 pW laser. We also investigated the photoresponsivity at the positive gate bias (+50 V), where the photoresponsivity

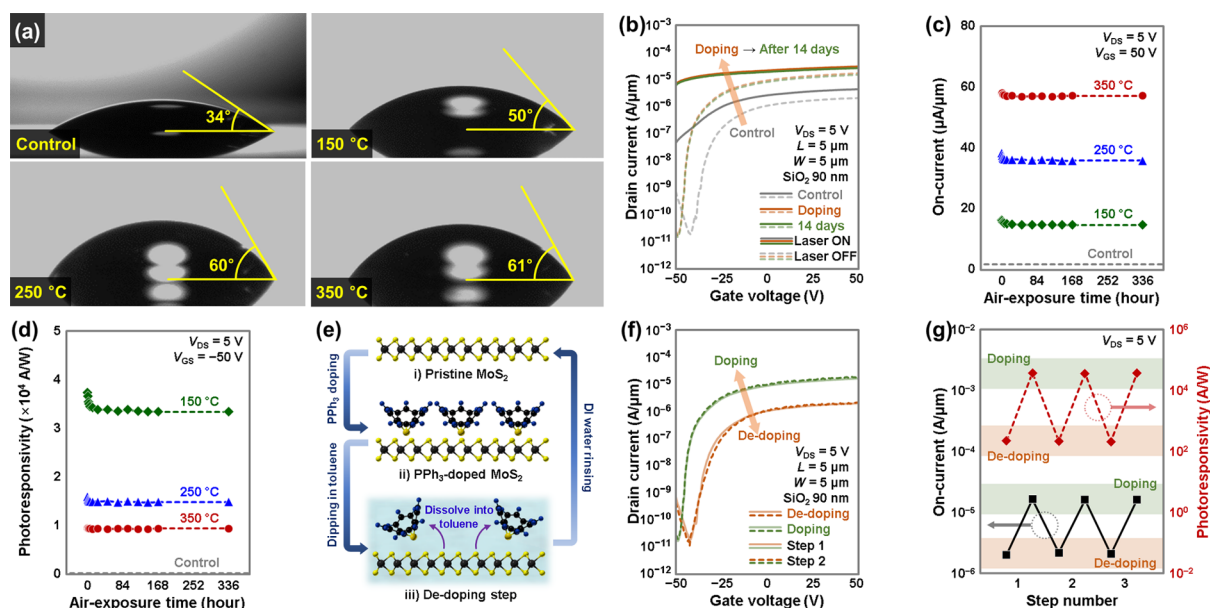


Figure 4. Stability and reusability of the PPh₃ doping process: (a) water contact angles measured on the surface of MoS₂ devices PPh₃ formed at three different temperatures (150, 250, and 350 °C); (b) transfer characteristics of the control and PPh₃-doped devices after 14 d of air exposure; (c) on-current and (d) photoresponsivity as a function of the air-exposure time, which were extracted from MoS₂ devices activated at different temperatures ($V_{DS} = 5$ V, and $V_{GS} = 50$ V for on-current and -50 V for photoresponsivity); (e) graphical depiction of the PPh₃ doping and de-doping processes; (f) reusability analysis of repeated PPh₃ doping and de-doping processes in terms of the I_D – V_G characteristics, where solid and dashed lines indicate drain currents under laser on and off conditions, respectively; (g) on-current and photoresponsivity.

was varied similarly to the case of negative gate bias condition (Supporting Information Figure S6). The extracted photoresponsivity values were comparable to those previously reported for MoS₂-based photodetectors (7.5×10^{-3} to 6.0×10^5 A/W)^{31–34} and were lower than those for ReS₂-based photodetectors (1.6×10^1 to 2.5×10^7 A/W),^{35,36} as summarized in Supporting Information Table S2. An inverse relationship existed between the photoresponsivity and incident laser power. This was because the scattering phenomenon in the photogenerated carriers was suppressed at lower illumination powers. Figure 3f shows the spectral responses of photodetectors exposed under an incident laser with a power of 1 nW. The responsivity of all devices decreased with increases in the wavelength from 520 to 850 nm because the photocurrent response is dependent on MoS₂ channel absorption and the amount of absorbed power is inversely proportional to the wavelength. The responsivity values exceeded 10^3 A/W in the selected wavelength region, thereby indicating that the photodetector can be potentially applied for various photonic device applications that require high photoresponsivity in the visible to near-infrared region. We have also characterized the photodetectivity (D^*) of the devices doped at three different temperatures. D^* decreased from 6.82×10^{13} to 2.36×10^{10} Jones as the PPh₃ formation temperature increased from 150 to 350 °C. This relationship between D^* and PPh₃ formation temperature is consistent with the photoresponsivity changing trend under -50 V gate bias. Further details are provided in Supporting Information Table S3. Additionally, temporal photoresponse characteristics were investigated in the control and PPh₃-doped MoS₂ devices with laser switching cycles of 20 s. Figure 3g shows the temporal photoresponse curves for the photodetectors doped with PPh₃ at different temperatures. The response time is related to the recombination rate of the photogenerated carriers, and thus each curve in Figure 3g exhibits different

rising and decaying behaviors. From the curves, the rising and decaying times (τ_r and τ_d) were extracted between 10 and 90% of the photocurrents, respectively. Both the rising and decaying times (τ_r and τ_d) increased slightly (τ_r : $6.87 \rightarrow 13.2$ s and τ_d : $12.7 \rightarrow 17.2$ s) with an increase in the PPh₃ formation temperature. Extracted values for all devices are listed in Supporting Information Table S4. The aforementioned slow temporal responses are potentially caused by the increases in the carrier lifetime with increases in the amount of tightly bound trions.

The water wettability of the PPh₃ layer on MoS₂ is a critical point that determines the stability of the PPh₃ doping and the doped devices. First, water contact angle analysis was performed for MoS₂ devices with PPh₃ layers formed at different temperatures (150, 250, and 350 °C), as shown in Figure 4a. The contact angle of the control MoS₂ was approximately 34°, but the angle increased up to 61° with the increases in the PPh₃ formation temperature. This indicates that the surface of PPh₃-doped MoS₂ was more hydrophobic than that of the control device. Because PPh₃ is an organic polymeric material composed of hydrophobic C–H (hydrocarbon) bonds, it is inherently water-insoluble and has a hydrophobic surface property. This PPh₃ layer became denser after annealing at a higher temperature, subsequently strengthening the hydrophobicity at the surface. The strengthened hydrophobicity makes it harder for moisture to attach to the channel surface and react with PPh₃ molecules. By virtue of the stronger hydrophobicity providing better surface protection, PPh₃ doping is expected to result in more stable characteristics with the increase in the formation temperature. We also note that the contact angle values measured in this work are lower than the previously reported values on MoS₂.³⁷ It is because we could measure 1 min after water was dropped on the surface owing to the calibration and focus times. Figure 4b shows the transfer characteristics under

laser on (solid line) and off (dashed line) conditions for the control and the PPh₃-doped devices immediately after doping and 14 d after doping. A significant performance degradation was not observed in the characteristic curves, thereby indicating that the PPh₃ doping exhibits excellent stability. With respect to devices doped with PPh₃ formed at lower temperatures (150 and 250 °C), the on-current level and photoresponsivity values decreased slightly after 336 h of exposure to air by 5.48–8.85 and 6.18–9.53%, respectively, as shown in Figure 4c,d. However, a significantly lower decrease was observed in the on-current (1.34%) and photoresponsivity (1.58%) after air-exposure for 336 h in the device doped at 350 °C. The superior stability is potentially attributable to the comparatively strong hydrophobicity predicted by the contact angle measurement, which is superior to that of previously reported doping methods for van der Waals materials.^{10,19} Additionally, every PPh₃-doped device maintained stable values in drain current level and photoresponsivity after an initial degradation time, as shown in Figure 4c,d. In addition to excellent air-stability, the controllable and reversible properties of the PPh₃ doping technique could provide very interesting and practical advantages. For example, it can be used to implement a channel region with different doping concentrations in one MoS₂ device through repetitive formation and removal of PPh₃ molecules. Figure 4e illustrates the PPh₃ doping and de-doping processes, where the PPh₃ molecules attached to the surface of MoS₂ could be easily removed in toluene because the PPh₃ solutions were prepared by dissolving triphenylphosphine in a solvent (toluene). Furthermore, the electrical performance of the MoS₂ device was not significantly affected by the repeated toluene-based de-doping processes. Figure 4f shows the I_D – V_G characteristics of a PPh₃-doped MoS₂ device during repeated doping and de-doping steps. In particular, we also investigated the doping effects on the hysteresis characteristic of the MoS₂ device in Supporting Information Figure S7 and Table S5. Despite the repeated doping processes, the degradation in the drain current and the change in the hysteresis gap were almost negligible. The variation in photoresponsivity and drain current during the repeated steps is summarized in Figure 4g. The current level remained constant with a variation of 0.99% after repeating the doping/de-doping process thrice and subsequently exhibited highly stable photoresponsivity.

CONCLUSIONS

In summary, a highly stable and reversible n-doping technique based on the PPh₃ molecule was developed in the study, which originated from negative charge transfer from the electron lone pair in the PPh₃ molecule. After PPh₃ n-doping, significant improvements in the electronic and optoelectronic performance were observed as follows: on/off-current ratio ($8.72 \times 10^4 \rightarrow 8.70 \times 10^5$); carrier mobility ($12.1 \rightarrow 241 \text{ cm}^2/\text{V}\cdot\text{s}$); and photoresponsivity ($2.77 \times 10^3 \rightarrow 3.92 \times 10^5 \text{ A/W}$). The doping concentration was broadly distributed between 1.56×10^{11} and $9.75 \times 10^{12} \text{ cm}^{-2}$ and was easily controlled by adjusting the temperature at which the PPh₃ layer was formed. In particular, the proposed doping technique provided two noteworthy advantages that have rarely been reported for previous molecular doping methods. First, the doping technique was highly stable, leading to very small variations in the on-current (1.34%) and photoresponsivity (1.58%) after exposure to air for 14 d. Second, the doping process was reversible and allowed for repetitive formation and removal of

PPh₃ molecules, thus enabling to reuse the devices. We believe that the proposed PPh₃ molecule-based doping technique can enable the realization of high-performance TMD devices and facilitate emerging applications that require feasible and reliable doping techniques.

EXPERIMENTAL METHODS

To prepare the doping solution with a PPh₃ concentration of 7.5 wt %, 75 mg of PPh₃ powder was dissolved in 925 mg of toluene. The prepared PPh₃ solution was spin-coated on MoS₂ at 1000 rpm for 30 s (5 s acceleration). The PPh₃-coated MoS₂ samples were subsequently annealed at three different temperatures (150, 250, and 350 °C) to activate the doping and remove any remaining solvent. The characteristics of PPh₃-doped MoS₂ were investigated using Raman and PL spectroscopy (Alpha 300 M+, WITec) with an excitation laser wavelength of 532 nm, laser beam size of 0.7–0.9 μm, and an instrumental spectral resolution less than 0.9 cm^{−1}. To fabricate the MoS₂ devices, MoS₂ flakes with approximately 45 layers were selected after mechanically exfoliating many flakes and then transferred to a 90 nm thick silicon dioxide layer on a heavily boron-doped silicon substrate via an adhesive tape. The source and drain electrode regions were subsequently patterned on the MoS₂ layer using a lithography process, followed by the deposition of 10 nm thick titanium and 50 nm thick gold layers with an e-beam evaporator. The fabricated transistors were investigated through current–voltage (I_D – V_G) measurements. The field-effect mobility (μ_{FE}) and carrier concentration (n) were extracted using the equations $\mu_{FE} = L/(WV_D C_{OX}) \times (\partial I_D / \partial V_G)$ and $n = I_D L / qW\mu_{FE} V_D$, where L and W denote the length and width of the channel, respectively, q denotes the electron charge, and C_{OX} denotes the gate capacitance per unit area. The fabricated photodetectors were also analyzed by performing electrical measurements under both dark and illuminated conditions. The light source was a dot laser with a wavelength of 520 nm and an optical power of 4.0 mW/cm². The active channel region of the photodetector is 25 μm² (5 μm × 5 μm), making the illuminated light power 1 nW. The photoresponsivity (R) was calculated using $R = I_{ph}/P_{light}$, where I_{ph} denotes the generated photocurrent and P_{light} denotes the total incident optical power that varied with values of 5, 10, 100, and 1000 pW by a mounted round variable ND filter. The temporal photoresponse was investigated under laser illumination that was turned on and off with a cycle time of 20 s. The rising and decaying times were extracted at between 10 and 90% of the measured maximum photocurrent.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.8b06767.

Mean and standard deviation values corresponding to E_{2g}^1 and A_{1g} peaks as extracted from mapping images; XPS spectra analyses of the P 2p and C 1s peaks; PL analysis of the optical characteristics of MoS₂; field-effect mobility extracted at low drain bias (0.1 V); contact resistance (R_c) at different doping conditions; photoresponsivity analysis under positive gate bias; comparison of the performance of photodetectors fabricated on TMDs and graphene; photodetectivity performance comparison; temporal photoresponse times (rising and decaying); and doping and de-doping effects on hysteresis characteristic (PDF)

AUTHOR INFORMATION

Corresponding Author

*E-mail: jhpark9@skku.edu.

ORCID

Seo-Hyeon Jo: 0000-0001-9075-011X

Jin-Hong Park: 0000-0001-8401-6920

Author Contributions

[§]K.H. and S.-H.J. contributed equally.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This research was supported by the Basic Science Research Program, Basic Research Lab Program, and Nano-Material Technology Development Program through National Research Foundation of Korea (NRF) grants funded by the Korea government (MSIP) (2018R1A2A2A05020475, 2015M3A7B7045496, 2017R1A4A1015400, and 2016M3A7B4910426), and the Future Semiconductor Device Technology Development Program (10067739) funded by Ministry of Trade, Industry and Energy (MOTIE), and Korea Semiconductor Research Consortium (KSRC).

■ REFERENCES

- (1) Mak, K. F.; Lee, C.; Hone, J.; Shan, J.; Heinz, T. F. Atomically Thin MoS₂: A New Direct-Gap Semiconductor. *Phys. Rev. Lett.* **2010**, *105*, 136805.
- (2) Ganatra, R.; Zhang, Q. Few-Layer MoS₂: A Promising Layered Semiconductor. *ACS Nano* **2014**, *8*, 4074–4099.
- (3) Zhao, W.; Ghorannevis, Z.; Chu, L.; Toh, M.; Kloc, C.; Tan, P.-H.; Eda, G. Evolution of Electronic Structure in Atomically Thin Sheets of WS₂ and WSe₂. *ACS Nano* **2013**, *7*, 791–797.
- (4) Li, H.; Wu, J.; Yin, Z.; Zhang, H. Preparation and Applications of Mechanically Exfoliated Single-Layer and Multilayer MoS₂ and WSe₂ Nanosheets. *Acc. Chem. Res.* **2014**, *47*, 1067–1075.
- (5) Chenet, D. A.; Aslan, O. B.; Huang, P. Y.; Fan, C.; van der Zande, A. M.; Heinz, T. F.; Hone, J. C. In-Plane Anisotropy in Mono- and Few-Layer ReS₂ Probed by Raman Spectroscopy and Scanning Transmission Electron Microscopy. *Nano Lett.* **2015**, *15*, S667–S672.
- (6) Lin, Y.-C.; Komsa, H.-P.; Yeh, C.-H.; Björkman, T.; Liang, Z.-Y.; Ho, C.-H.; Huang, Y.-S.; Chiu, P.-W.; Krashenninnikov, A. V.; Suenaga, K. Single-Layer ReS₂: Two-Dimensional Semiconductor with Tunable In-Plane Anisotropy. *ACS Nano* **2015**, *9*, 11249–11257.
- (7) Zhang, E.; Jin, Y.; Yuan, X.; Wang, W.; Zhang, C.; Tang, L.; Liu, S.; Zhou, P.; Hu, W.; Xiu, F. ReS₂-Based Field-Effect Transistors and Photodetectors. *Adv. Funct. Mater.* **2015**, *25*, 4076–4082.
- (8) Fang, H.; Chuang, S.; Chang, T. C.; Takei, K.; Takahashi, T.; Javey, A. High-Performance Single Layered WSe₂ p-FETs with Chemically Doped Contacts. *Nano Lett.* **2012**, *12*, 3788–3792.
- (9) Zhao, P.; Kiriya, D.; Azcatl, A.; Zhang, C.; Tosun, M.; Liu, Y.-S.; Hettick, M.; Kang, J. S.; McDonnell, S.; KC, S.; Guo, J.; Cho, K.; Wallace, R. M.; Javey, A. Air Stable p-Doping of WSe₂ by Covalent Functionalization. *ACS Nano* **2014**, *8*, 10808–10814.
- (10) Kang, D.-H.; Kim, M.-S.; Shim, J.; Jeon, J.; Park, H.-Y.; Jung, W.-S.; Yu, H.-Y.; Pang, C.-H.; Lee, S.; Park, J.-H. High-Performance Transition Metal Dichalcogenide Photodetectors Enhanced by Self-Assembled Monolayer Doping. *Adv. Funct. Mater.* **2015**, *25*, 4219–4227.
- (11) Park, H.-Y.; Lim, M.-H.; Jeon, J.; Yoo, G.; Kang, D.-H.; Jang, S. K.; Jeon, M. H.; Lee, Y.; Cho, J. H.; Yeom, G. Y.; Jung, W.-S.; Lee, J.; Park, S.; Lee, S.; Park, J.-H. Wide-Range Controllable n-Doping of Molybdenum Disulfide (MoS₂) through Thermal and Optical Activation. *ACS Nano* **2015**, *9*, 2368–2376.
- (12) Shim, J.; Oh, A.; Kang, D.-H.; Oh, S.; Jang, S. K.; Jeon, J.; Jeon, M. H.; Kim, M.; Choi, C.; Lee, J.; Lee, S.; Yeom, G. Y.; Song, Y. J.; Park, J.-H. High-Performance 2D Rhenium Disulfide (ReS₂) Transistors and Photodetectors by Oxygen Plasma Treatment. *Adv. Mater.* **2016**, *28*, 6985–6992.
- (13) Lin, J. D.; Han, C.; Wang, F.; Wang, R.; Xiang, D.; Qin, S.; Zhang, X.-A.; Wang, L.; Zhang, H.; Wee, A. T. S.; Chen, W. Electron-doping-enhanced Trion Formation in Monolayer Molybdenum Disulfide Functionalized with Cesium Carbonate. *ACS Nano* **2014**, *8*, 5323–5329.
- (14) Kiriya, D.; Tosun, M.; Zhao, P.; Kang, J. S.; Javey, A. Air-Stable Surface Charge Transfer Doping of MoS₂ by Benzyl Viologen. *J. Am. Chem. Soc.* **2014**, *136*, 7853–7856.
- (15) Nam, H.-J.; Kim, J.; Park, J.-H. Wide-Range Controllable Doping of Tungsten Diselenide (WSe₂) based on Hydrochloric Acid Treatment. *J. Phys. Chem. C* **2017**, *121*, 14367–14372.
- (16) Lei, S.; Wang, X.; Li, B.; Kang, J.; He, Y.; George, A.; Ge, L.; Gong, Y.; Dong, P.; Jin, Z.; Brunetto, G.; Chen, W.; Lin, Z.-T.; Baines, R.; Galvão, D. S.; Lou, J.; Barrera, E.; Banerjee, K.; Vajtai, R.; Ajayan, P. Surface functionalization of two-dimensional metal chalcogenides by Lewis acid-base chemistry. *Nat. Nanotechnol.* **2016**, *11*, 465–471.
- (17) Denis, P. A. Chemical Reactivity and Band-gap Opening of Graphene Doped with Gallium, Germanium, Arsenic, and Selenium Atoms. *ChemPhysChem* **2014**, *15*, 3994–4000.
- (18) Fathipour, S.; Pandey, P.; Fullerton-Shirey, S.; Seabaugh, A. Electric-double-layer doping of WSe₂ field-effect transistors using polyethylene-oxide cesium perchlorate. *J. Appl. Phys.* **2016**, *120*, 234902.
- (19) Gobbi, M.; Orgiu, E.; Samori, P. When 2D Materials Meet Molecules: Opportunities and Challenges of Hybrid Organic/Inorganic Van der Waals Heterostructures. *Adv. Mater.* **2018**, *30*, 1706103.
- (20) Yang, J.; Kim, S.; Choi, W.; Park, S. H.; Jung, Y.; Cho, M.-H.; Kim, H. Improved Growth Behavior of Atomic-Layer-Deposited High-k Dielectrics on Multilayer MoS₂ by Oxygen Plasma Pretreatment. *ACS Appl. Mater. Interfaces* **2013**, *5*, 4739–4744.
- (21) Rai, A.; Valsaraj, A.; Movva, H. C. P.; Roy, A.; Ghosh, R.; Sonde, S.; Kang, S.; Chang, J.; Trivedi, T.; Dey, R.; Guchhait, S.; Larentis, S.; Register, L. F.; Tutuc, E.; Banerjee, S. K. Air Stable Doping and Intrinsic Mobility Enhancement in Monolayer Molybdenum Disulfide by Amorphous Titanium Suboxide Encapsulation. *Nano Lett.* **2015**, *15*, 4329–4336.
- (22) Park, H.-Y.; Dugasani, S. R.; Kang, D.-H.; Jeon, J.; Jang, S. K.; Lee, S.; Roh, Y.; Park, S. H.; Park, J.-H. n- and p-type Doping Phenomenon by Artificial DNA and M-DNA on Two-dimensional Transition Metal Dichalcogenides. *ACS Nano* **2014**, *8*, 11603–11613.
- (23) Kang, D.-H.; Dugasani, S. R.; Park, H.-Y.; Shim, J.; Gnappareddy, B.; Jeon, J.; Lee, S.; Roh, Y.; Park, S. H.; Park, J.-H. Ultra-low n- and p-Doping on Two-dimensional Transition Metal Dichalcogenides using DNA Nanostructure doped by Combination of Lanthanide and Metal Ions. *Sci. Rep.* **2016**, *6*, 20333.
- (24) Parkin, W. M.; Balan, A.; Liang, L.; Das, P. M.; Lamparski, M.; Naylor, C. H.; Rodríguez-Manzo, J. A.; Johnson, A. T. C.; Meunier, V.; Drndić, M. Raman Shifts in Electron-Irradiated Monolayer MoS₂. *ACS Nano* **2016**, *10*, 4134–4142.
- (25) Jo, S.-H.; Kang, D.-H.; Shim, J.; Jeon, J.; Jeon, M. H.; Yoo, G.; Kim, J.; Lee, J.; Yeom, G. Y.; Lee, S.; Yu, H.-Y.; Choi, C.; Park, J.-H. A High-Performance WSe₂/h-BN Photodetector using a Triphenylphosphine (PPh₃)-Based n-Doping Technique. *Adv. Mater.* **2016**, *28*, 4824–4831.
- (26) Fang, H.; Tosun, M.; Seol, G.; Chang, T. C.; Takei, K.; Guo, J.; Javey, A. Degenerate n-Doping of Few-Layer Transition Metal Dichalcogenides by Potassium. *Nano Lett.* **2013**, *13*, 1991–1995.
- (27) Tarasov, A.; Zhang, S.; Tsai, M.-Y.; Campbell, P. M.; Graham, S.; Barlow, S.; Marder, S. R.; Vogel, E. M. Controlled Doping of Large-Area Trilayer MoS₂ with Molecular Reductants and Oxidants. *Adv. Mater.* **2015**, *27*, 1175–1181.
- (28) Najmaei, S.; Zou, X.; Er, D.; Li, J.; Jin, Z.; Gao, W.; Zhang, Q.; Park, S.; Ge, L.; Lei, S.; Kono, J.; Shenoy, V. B.; Yakobson, B. I.; George, A.; Ajayan, P. M.; Lou, J. Tailoring the Physical Properties of Molybdenum Disulfide Monolayers by Control of Interfacial Chemistry. *Nano Lett.* **2014**, *14*, 1354–1361.
- (29) Chen, K.; Kiriya, D.; Hettick, M.; Tosun, M.; Ha, T.-J.; Madhvapathy, S. R.; Desai, S.; Sachid, A.; Javey, A. Air stable n-doping

of WSe₂ by Silicon Nitride Thin Films with Tunable Fixed Charge Density. *APL Mater.* **2014**, *2*, 092504.

(30) Ali, M. H.; Kang, D.-H.; Park, J.-H. Rhenium diselenide (ReSe₂) infrared photodetector enhanced by (3-aminopropyl)-trimethoxysilane (APTMS) treatment. *Org. Electron.* **2018**, *53*, 14–19.

(31) Lopez-Sanchez, O.; Lembke, D.; Kayci, M.; Radenovic, A.; Kis, A. Ultrasensitive photodetectors based on monolayer MoS₂. *Nat. Nanotechnol.* **2013**, *8*, 497–501.

(32) Kang, D.-H.; Kim, M.-S.; Shim, J.; Jeon, J.; Park, H.-Y.; Jung, W.-S.; Yu, H.-Y.; Pang, C.-H.; Lee, S.; Park, J.-H. High-Performance Transition Metal Dichalcogenide Photodetectors Enhanced by Self-Assembled Monolayer Doping. *Adv. Funct. Mater.* **2015**, *25*, 4219–4227.

(33) Kufer, D.; Nikitskiy, I.; Lasanta, T.; Navickaite, G.; Koppens, F. H. L.; Konstantatos, G. Hybrid 2D-0D MoS₂-PbS Quantum Dot Photodetectors. *Adv. Mater.* **2015**, *27*, 176–180.

(34) Kang, D.-H.; Pae, S. R.; Shim, J.; Yoo, G.; Jeon, J.; Leem, J. W.; Yu, J. S.; Lee, S.; Shin, B.; Park, J.-H. An Ultrahigh-Performance Photodetector based on a Perovskite-Transition-Metal-Dichalcogenide Hybrid Structure. *Adv. Mater.* **2016**, *28*, 7799–7806.

(35) Choi, W.; Cho, M. Y.; Konar, A.; Lee, J. H.; Cha, G.-B.; Hong, S. C.; Kim, S.; Kim, J.; Jena, D.; Joo, J.; Kim, S. High-Detectivity Multilayer MoS₂ Phototransistors with Spectral Response from Ultraviolet to Infrared. *Adv. Mater.* **2012**, *24*, 5832–5836.

(36) Shim, J.; Oh, A.; Kang, D.-H.; Oh, S.; Jang, S. K.; Jeon, J.; Jeon, M. H.; Kim, M.; Choi, C.; Lee, J.; Lee, S.; Yeom, G. Y.; Song, Y. J.; Park, J.-H. High-Performance 2D Rhenium Disulfide (ReS₂) Transistors and Photodetectors by Oxygen Plasma Treatment. *Adv. Mater.* **2016**, *28*, 6985–6992.

(37) Gaur, A. P. S.; Sahoo, S.; Ahmadi, M.; Dash, S. P.; Guinel, M. J.-F.; Katiyar, R. S. Surface Energy Engineering for Tunable Wettability through Controlled Synthesis of MoS₂. *Nano Lett.* **2014**, *14*, 4314–4321.